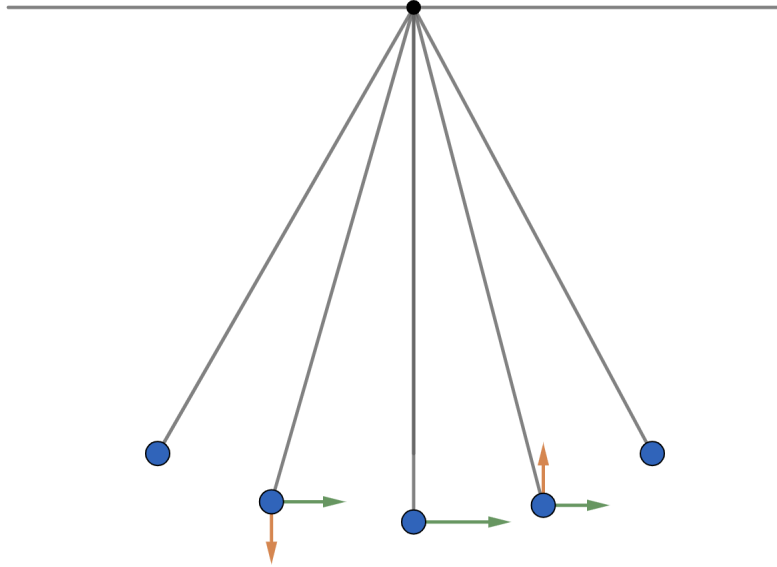


2023 F=ma Exam: Problem 20

Kevin S. Huang



Taking the initial state as the pendulum starting on the left, we have $(v_x, v_y) = (0, 0)$. As it swings towards the center, v_x is always increasing while v_y decreases and then increases back to 0.

Going from the center to the right side, v_x is always decreasing until it goes back to 0 while v_y increases and then decreases back to 0. We are back at $(v_x, v_y) = (0, 0)$.

Going back from the right side to the left side, everything is the same except v_x flips direction compared to before. This corresponds to reflecting the earlier closed trajectory across the y-axis on the v_x - v_y graph.

The graph C depicts the described motion.